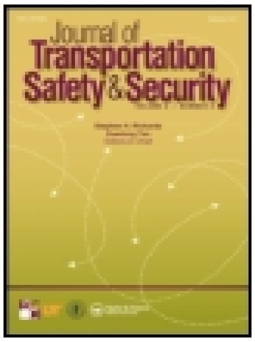

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Examining two-wheelers' overtaking behavior and lateral distance choices at a shared roadway facility

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ABSTRACT

This article investigates the lateral distance that overtaking two-wheelers (bicycles, e-bikes, and e-scooters) keep from automobiles at shared traffic streets. A video-based computer vision technique is used to track road users, collect their trajectories, and measure the lateral distance. A full Bayesian logit model is developed to examine the factors that affect the likelihood of two-wheelers accepting the critical lateral distance that is defined as the 10th percentile lateral distance. The results show that (a) the average lateral distance between overtaking two-wheelers and automobiles is 1.54 m, (b) the lateral distance for bicycles is significantly larger than that for e-bikes and e-scooters, (c) the lateral distance follows a best-fitted Gamma distribution. Further results from the full Bayesian logit model show that (a) two-wheelers type, evasive action manner, occurrence of a platoon of moving two-wheelers, and two-wheelers' yaw rate ratio are significantly positively related to the probability of two-wheelers accepting the critical lateral distance and (b) the presence of heavy vehicles and the speed difference between two-wheelers and interacting automobiles are negatively associated with the above probability.

KEYWORDS

shared roadway; lateral distance; logit model; full Bayesian; computer vision

1. Introduction

Two-wheelers (including bicycles, e-bikes, and e-scooters) are increasingly promoted as a sustainable urban mode of transportation (Guo, Liu, Bai, Xu, & Chen, 2014). They make an efficient use of road resources and have a significant impact on reducing traffic congestion, energy consumption, and air pollution (Li, Wang, Liu, & Ragland, 2012). Due to the limited road space, two-wheelers usually share the road space with automobiles (including passenger cars and heavy vehicles). They are more flexible and can take advantage of their small size to navigate through traffic by overtaking whereas automobiles usually are confined in their lanes. The overtaking behavior of two-wheelers may lead to complex lateral interactions

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between two-wheelers and automobiles. These lateral interactions can result in a high potential of traffic collisions (Tageldin, Sayed, & Wang, 2015), which have a significant influence on traffic performance and safety (Sayed, Zaki, & Autey, 2013). In fact, two-wheelers–involved crashes are usually over-represented in traffic collisions at shared roadways (Naci, Chisholm, & Baker, 2009). Among the two-wheelers–involved crashes, collisions during overtaking maneuvers are cited as one of the primary causes of two-wheelers fatalities (Kay, Savolainen, Gates, & Datta, 2014; Transport for London, 2005). Considering these issues, there is a need to perform thorough analysis of the lateral interactions between overtaking two-wheelers and automobiles.

Traditionally, information related to lateral distance and interactions between two-wheelers and automobiles has been collected by an instrumented bicycle (Chuang, Hsu, Lai, Doong, & Jeng, 2013; Mehta, Mehran, & Hellinga, 2015; Shackel & Parkin, 2014; Walker, 2007). The bicycle is equipped with an ultrasonic distance sensor, Global Positioning System (GPS) receiver, microcontroller, and data logger. However, this approach has shortcomings as it can be unreliable and costly. As well, because of the small samples (usually only one instrumented bicycle in the data collection process), the lateral distance analysis may result in a biased evaluation. Simulation models such as the car-following and cellular automata models are also used for lateral distance data collection (Meng, Dai, Dong, & Zhang, 2007). This approach inherits the limitations typically associated with simplified models, difficulty of parameter calibration, and the challenges to replicate the drivers' complex behavior. Recently, computer vision analysis is being promoted as a leading technology for traffic analysis. It enables data collection of road users at a large scale and granular details necessary to capture their behavior (Guo, Sayed, & Zaki, 2016, 2018; Laureshyn, Ardö, Svensson, & Jonsson, 2009; Zaki & Sayed, 2016).

The objective of this study is to examine the lateral distance between overtaking two-wheelers and automobiles at shared traffic streets with weak lane-discipline environment. The *lateral distance* in this study refers to the distance that overtaking two-wheelers keep from automobiles (the two-wheelers decide the lateral distance). The following contributions are presented: (1) to compare the lateral distance between automobiles and different types of two-wheelers in overtaking maneuver, (2) to analyze the distribution of the lateral distance, and (3) to examine the factors that affect the likelihood of two-wheelers accepting the critical lateral distance. Video data are collected from a middle-block shared traffic street. Traffic data and road-users trajectories are automatically collected using video-based computer vision technique.

The research can be beneficial for development of countermeasures focusing on two-wheelers safety issues. Furthermore, it can be used to improve the microsimulation models in mixed traffic environment.

2. Literature review

2.1. Lateral distance between automobiles and two-wheelers

Previous studies have been performed to investigate the lateral interactions between motor vehicles and bicycles/motorcycles, particularly focusing on the lateral distance analysis. Walker (2007) employed a naturalistic experiment to collect proximity data from motorists overtaking bicycles. They found that overtaking motorized vehicles tended to provide more lateral space to bicyclists perceived as females than males and less space when bicyclists were wearing helmets. It was also found that the passing large vehicles were particularly likely to leave narrow lateral space to bicyclists.

Parkin and Meyers (2010) investigated the lateral distance between motor vehicles and cyclists on highways with and without bicycle lanes using the data collected from an instrumented bicyclist. The study showed that vehicle drivers provided less lateral distance for cyclists in the presence of a bicycle lane. A similar study was performed by Mehta et al. (2015) that analyzed the influence of on-street bike lanes on the lateral separation between automobiles and cyclists. The result showed that the facilities with on-street bike lanes provided greater separation between bicycles and automobiles. It also showed that on arterial roadways without on-street bike lanes drivers tended to provide more lateral clearance to overtaken bicycles by either changing lanes or by encroaching on the adjacent lane.

Sando, Chimba, Kwigizile, and Moses (2011) used multivariate regression to analyze the lateral distance between motorized vehicles and bicycles. The result indicated that vehicles pass closer to cyclists when a cyclist is a male, traffic volume is high, curb lane is narrower, or when passing is done by small vehicles. Chuang et al. (2013) utilized an instrumented bicycle to investigate the factors that affect the motorists' initial lateral passing distance and the bicyclists' position (lateral distance from the passing motorists). The study showed that lateral distance during passing events was smaller for motorcycles than for cars and small trucks. It was also found that motor vehicles tended to give more space to female bicyclists than males. Kay et al. (2014) found that the vehicle type, cyclist position, the presence of centerline rumble strips, and the opposing traffic have significant impact on the lateral distance of automobiles passing cyclists.

Shackel and Parkin (2014) examined the influence of road markings, lane widths, and driver behavior on proximity and speed of vehicles overtaking cyclists. The results suggested that the road width has a positive effect on the distance of motorized vehicle passing cyclists whereas the nearside parking and an opposing vehicle reduce the passing distance given by motorized vehicles. Stewart and McHale (2014) developed two generalized linear model (GLM) models for the passing distance afforded to bicycles on urban roads. Several factors were found to be related to the passing distance including opposing automobiles, effective lane width, bicycle lane width, bicycle and vehicle speed, nearside and opposite side parking, and the presence of traffic island.

Zhao, Guo, Wang, and Jiang (2015) used a survival analysis to model the lateral interferences between motor vehicles and nonmotor vehicles. It was observed that drivers are inclined to keep larger lateral distance to nonmotor vehicles. The nonmotor vehicle drivers accept shorter lateral distances than the motor vehicle drivers. Using the data collected from six shared road segments by video cameras, Apasnore, Ismail, and Kassim (2017) examined the bicycle–vehicle interaction at midsection of mixed traffic street at Ottawa, Canada. The result showed that 90% of passes exceed 1.23 m and the passing distance is positively correlated with automobiles speed, lane width, and bicycle position from adjacent curb edge line, while inversely correlated to ambient traffic density and bicycle speed.

In general, few studies in the literature investigated the lateral distance of two-wheelers overtaking motor vehicles. Vlahogianni (2014) evaluated the powered-two-wheelers (PTWs) kinematic characteristics and their interactions with motorized vehicles, as well as the factors that may affect the likelihood of PTW drivers to accept critical spacing during filtering and overtaking. The result indicated that the kinematic parameters for the patterns of filtering and overtaking have several differences. Barmpounakis, Vlahogianni, and Golias (2016) utilized game theory to model the cooperation between PTW drivers and other drivers during PTWs' overtaking maneuver. Findings of this study indicated that the successful overtaking rate is higher when the PTW driver is noncooperative, whereas lower overtaking rates occur when the driver of the lead vehicle is noncooperative. Sugandhi and Chunchu (2017) modeled the path selection behavior of a motorized two-wheeler (MTW) overtaking a light motor vehicle (LMV). It was found that the width available on a particular path for overtaking and the headway difference with the lead and lag vehicles on the target paths significantly affected the decision of the MTW in choosing a particular path.

2.2. The current article

The review of previous studies showed that most of the research examined the lateral distance of automobiles overtaking bicycles/motorcycles.

However, few studies focused on the lateral distance of two-wheelers overtaking automobiles. Previous studies have relatively limited capacity to explain the characteristics of lateral distance that two-wheelers keep from automobiles during two-wheelers' overtaking maneuver. In addition, little is known about how this lateral distance can vary among different types of the overtaking two-wheelers such as bicycles, e-bikes, and e-scooters which are emerging transport modes around the world. This study contributes to the literature by exploring and comparing the lateral distance between automobiles and different types of overtaking two-wheelers. The underlying factors that affect the likelihood of two-wheelers accepting the critical lateral distance were also examined using an advanced statistical method.

3. Data collection

3.1. Study location

The map-based random sampling and on-site visit were used to select the study location. Firstly, several potential locations in Kunming, China, which were used in previous studies (Bai, Liu, Guo, & Yu, 2015; Guo et al., 2014), were selected. These potential locations have high volume of two-wheelers and automobiles as well as heavy conflicts between them. Then, the Baidu-street view map was used to check if these roads meet the physical criteria (i.e., no segregation between bicycle lanes and motor lanes). Lastly, the sites were visited and the traffic was observed for 15 minutes at peak hour to confirm the presence of lateral interactions between automobiles and two-wheelers during overtaking events. The possibility to install video cameras for unobtrusive observation was also check during on-site visit.

The study location selected for the study is a middle-block shared traffic street. A camera is mounted on a pole by the traffic management authorities. Video data were recorded of the location for 1 ½ hours; a sample data that is considered adequate given the high-traffic volume of both two-wheelers and automobiles.

Figure 1 shows the layout of the study area. The street is a major east–west road with three lanes per direction. The outside lane is a shared bicycle lane for e-bikes, e-scooters, and bicycles, whereas the inside lanes are designed for automobiles. However, field observations show many road users being engaged in frequent illegal lane changing behavior due to lack of compliance with traffic regulations. E-bikes and e-scooters frequently change from their dedicated lane to the motor lane for pursuing higher speed. As such, this block has numerous lateral interactions. The study area focused on the west–east direction.

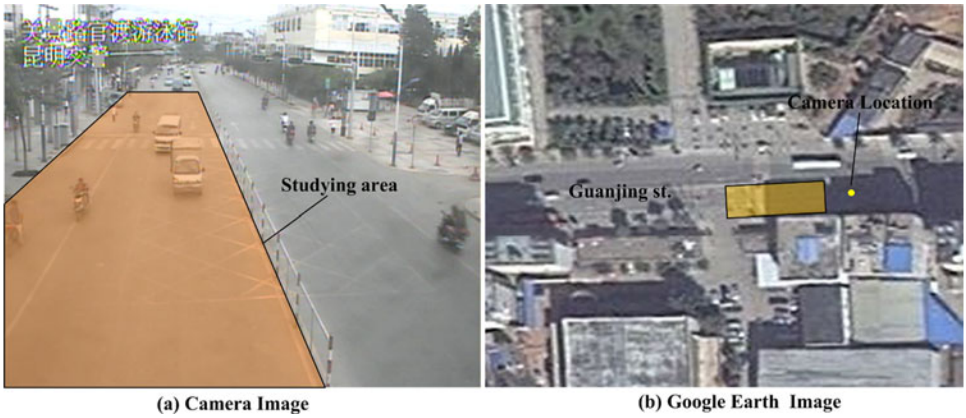

Figure 1. Layout of the study area on the shared traffic street.

3.2. Road-users tracking and lateral distance collection

Video data is processed using computer vision to extract the spatial-temporal information of each road user (Guo et al., 2017; Saunier, Sayed, & Ismail, 2010; Saunier & Sayed, 2006). In the first step of video processing, features are tracked on moving road users using the Kanade–Lucas–Tomasi (KLT) Feature Tracker algorithm (Saunier & Sayed, 2006) (see Figure 2a). A camera calibration procedure relates the tracked trajectories in the video image coordinates to their actual positions in the real world by creating a mapping between the three-dimensional real world and the two-dimensional image space (Ismail, Sayed, & Saunier, 2013) (see Figure 2b). In the next step, features that move at comparable speeds and fulfill defined spatial and motion constraints are grouped to create coherent objects.

To ensure a more accurate information about the lateral distance between the road users, information about their geometry would be necessary. An improvement of the object grouping is developed in this article by associating a box-shaped model to each detected road user (see Figure 2c). The detection of road users is enhanced using a 3-D bounding box representation. This is a reasonable assumption given the form of the road users which can be represented in rectangular shapes in the three-dimensional universe. By detailing the road users boundaries through the 3-D model, it is feasible to identify the corners of the road users. The process of the bounding box is invoked at the end of the object grouping process. The tracking accuracy for road users has been measured on several data sets to be between 84.7% and 94.4% (Saunier & Sayed, 2006).

The tracking precision is obtained from the feature tracking and grouping validation and is dependent on the ability to generate a movement trajectory for road users. The 3-D box is generated to bound the space occupied by each road user at each time instant. The peripheral of the box is limited by the location of corner features and encloses all the features

(a) Features Tracking

(b) Camera Calibration

(c) Objects 3-D Grouping

Figure 2. Road-users tracking and grouping.

associated with that object. By construction, the box accuracy is dependent on two factors. First, it features coverage. If the features from the tracking step are not evenly distributed on the road users, then the bounding box can underapproximate the region occupied by the road user. Second is camera calibration. Overapproximation of the bounding box can occur if the camera calibration accuracy is not satisfactory. This inaccuracy leads to "stretching" the distance between the top and lower features on a road user, and the bounding box appears larger than the region occupied by the road user. We took specific measures to reduce both problems. We ensured during the tracking phase that the quality and density of features covered the different parts of the road users. As such, the accuracy of the bounding box was directly related to the camera calibration. During the camera calibration process, we increased the number of road marks to decrease the positional errors.

A trajectory-based classification is then applied to identify the type of each road user. Characteristics that describe the geometry of the road users, such as the estimated length and width of the region occupied by the road user, are used to select those particular road users. As such, vehicles and pedestrians are identified from the tracked objects. The road users movement features such as maximum speed and movement pattern are used as cues to distinguish bicycles and MTWs. Specifically, a cyclist will generate a movement profile with pedaling process variations. In such case, the cadence (pedaling frequency) can be a good indicator that the observed trajectory belongs to a cyclist. On the other hand, MTWs movement patterns are primarily composed of linear segments. Once the set of characteristics are collected and their values estimated for each road user, a classification algorithm assigns a class label for each road user. In the end, the road users are classified into automobiles, MTWs, bicycles, and pedestrians. Ground truth validation, which is a comparison between the manual classified objects with the automated classified objects, was conducted to validate the classification result. The accuracy for road-user classification is above 90%. Furthermore, the missclassified objects were corrected manually. Subsequently, the MTWs were classified into e-bikes and e-scooters manually. A comprehensive description of the classification process can be found in Zaki, Sayed, and Wang (2016) and Zaki and Sayed (2013).

The lateral distance between two road users is estimated by calculating, at each time frame, the minimum distance between the two road users based on the minimum distance between the associated bounding boxes. As such, it is possible to analyze the variation in the lateral distance between the road users and also identify the time instant when the road users are closest to each other. Only the lateral distance between overtaking two-wheelers and automobiles are kept for further analysis in this study. Figure 3 shows the illustration of the lateral distance between two-wheelers and automobiles.

4. Methodology

4.1. Full Bayesian binary logit model

In this study, a full Bayesian binary logit (FBBL) model was developed to investigate factors associated with two-wheelers accepting the critical lateral distance. In contrast to the classical likelihood-based approach, the full Bayesian approach assuming the parameters as random variables and incorporates the uncertainty of the parameters through the posterior distribution. As such, the full Bayesian approach can avoid the overfitting problem and improve the predicting accuracy (Huber & Train, 2001; Guo, Li, Wu, & Xu, 2018; Xu, Wang, Liu, & Zhang, 2015). The FBBL model offers a flexible structure that allows the parameters following a posterior distribution. It can be written as:

$$Y_i \sim \text{Bernoulli}\ (p_i) \tag{1}$$

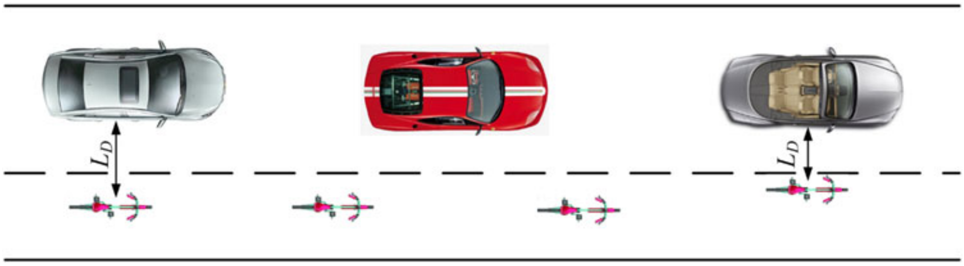

Figure 3. Lateral distance between two-wheelers and automobiles.

$$\text{logit}(p_i) = \ln\left[\frac{p_i}{1-p_i}\right] = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} \tag{2}$$

The Equation (2) can be rewritten as:

$$P(Y_i = 1) = \frac{\exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki})}{1 + \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki})} \tag{3}$$

Where, Y_i represents two-wheelers accepting the critical lateral distance (yes $= 1$, otherwise $= 0$) for the ith observation; p_i denotes the probability of accepting the critical lateral distance for the ith observation; x_{ki} denotes the value of explanatory variable k for for the ith observation; and β_k is the coefficient of explanatory variable k. The likelihood function can be written as:

$$\begin{aligned} f(Y|\boldsymbol{\beta}) &= \prod_{i=1}^{N}\left[(P(Y_i=1))^{y_i}(1-P(Y_i=1))^{1-y_i}\right] \\ &= \prod_{i=1}^{N}\left[\left(\frac{\exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki})}{1 + \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki})}\right)^{y_i}\right. \\ &\quad \left.\times\left(1 - \frac{\exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki})}{1 + \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki})}\right)^{(1-y_i)}\right] \end{aligned} \tag{4}$$

A Bayesian inference approach based on Markov Chain Monte Carlo (MCMC) method was adopted to simulate the posterior distribution of the coefficient vector $\boldsymbol{\beta}$ (i.e. $\boldsymbol{\beta} = [\beta_0, \beta_1, \ldots, \beta_k]^T$). The estimates of the Mean, Standard Deviation, and quartiles of the parameters of each explanatory variable can be determined by the posterior distribution provided by the Bayesian approach. Based on the specification of the Bayes theorem, the posterior distribution of parameters can be estimated using the following function

$$f(\boldsymbol{\beta}|\boldsymbol{Y}) = \frac{f(\boldsymbol{Y},\boldsymbol{\beta})}{f(\boldsymbol{Y})} = \frac{f(\boldsymbol{Y}|\boldsymbol{\beta})\pi(\boldsymbol{\beta})}{\int f(\boldsymbol{Y},\boldsymbol{\beta})d\boldsymbol{\beta}} \propto f(\boldsymbol{Y}|\boldsymbol{\beta})\pi(\boldsymbol{\beta}) \tag{5}$$

Where, $f(\boldsymbol{\beta}|\boldsymbol{Y})$ denotes the posterior joint distribution of parameters $\boldsymbol{\beta}$ conditional on data set $\boldsymbol{Y}$. $f(\boldsymbol{Y}, \boldsymbol{\beta})$ represents the joint probability distribution of data set $\boldsymbol{Y}$ and model parameters $\boldsymbol{\beta}$. $f(\boldsymbol{Y}|\boldsymbol{\beta})$ is the likelihood conditional function on parameters $\boldsymbol{\beta}$, specified by the Equation (4). Function $\pi(\boldsymbol{\beta})$ is the prior distribution of parameters $\boldsymbol{\beta}$. The following noninformative prior distributions were used in this study

$$\beta \sim N\left(0_k,\ 10^6 I_k\right) \tag{6}$$

Where, $\mathbf{0}_k$ is $k \times 1$ vector of zeros; $\boldsymbol{I}_k$ is a $k \times k$ identity matrix.

Based on the specification of the prior distributions for parameters $\boldsymbol{\beta}$, the posterior joint distribution $f(\boldsymbol{\beta}|\boldsymbol{Y})$ can be derived as

$$\begin{aligned} f(\boldsymbol{\beta}|\boldsymbol{Y}) &\propto f(\boldsymbol{Y}|\boldsymbol{\beta})\pi(\boldsymbol{\beta}) \\ &= \prod_{i=1}^{N}\left[\left(\frac{\exp\left(\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_k x_{ki}\right)}{1+\exp\left(\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_k x_{ki}\right)}\right)^{y_i}\right. \\ &\quad \left.\times\left(1-\frac{\exp\left(\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_k x_{ki}\right)}{1+\exp\left(\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_k x_{ki}\right)}\right)^{(1-y_i)}\right] \\ &\quad \times\prod_{i=1}^{N}\left[\frac{1}{\sqrt{2\pi}10^3}\exp\left(-\frac{1}{2}\frac{(\beta_k)^2}{10^6}\right)\right] \\ &\propto \exp\left\{\sum_{i=1}^{N}(\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_k x_{ki})y_i\right. \\ &\quad -\sum_{i=1}^{N}\log\left[1+e^{\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_k x_{ki}}\right] \\ &\quad \left.-\frac{1}{2}\frac{(\beta_0)^2+(\beta_1)^2+(\beta_2)^2+\cdots+(\beta_k)^2}{10^6}\right\} \end{aligned} \tag{7}$$

To generate realizations from the posterior joint distribution $f(\boldsymbol{\beta}|\boldsymbol{Y})$, the Metropolis–Hasting sampling approach sequentially draws parameters from Equation (7). The inference can then be made on the basis of the remaining draws after discarding the draws during the burn-in period.

4.2. Odds ratio analysis

The odds ratio (OR) is used to quantify the impacts of explanatory variables on the outcome. The OR is calculated for variables of interest in the FBBL model. The OR of an explanatory variable represents the increase in the odds of the outcome if the value of the variable increases by one unit

(Washington, Karlaftis, & Mannering, 2003). The OR for a variable x_j in the FBBL model can be calculated as

$$\mathrm{OR} = \frac{\mathrm{odds}(\boldsymbol{X}, x_j + 1)}{\mathrm{odds}(\boldsymbol{X}, x_j)} = \frac{\exp(\boldsymbol{X\beta}) \times \exp(\beta_j)}{\exp(\boldsymbol{X\beta})} = \exp(\beta_j) \tag{8}$$

5. Results and discussion

5.1. Lateral distance comparison

A total of 352 overtaking events were detected and extracted from the video using the computer vision technique. The lateral distance between overtaking two-wheelers and automobiles are collected automatically. The lateral distance varies from 0.33 m to 2.49 m with a mean of 1.542 m. The average lateral distance in this study is slightly higher than that in Vlahogianni (2014) (i.e., 1.542 m vs. 1.31 m), but slightly lower than that in (Zhao et al., 2015) (i.e., 1.542 m vs. 1.68 m). To compare the lateral distance between automobiles and different types of overtaking two-wheelers, two-wheelers are categorized into e-bikes, e-scooters, and bicycles. Figure 4 shows that the average lateral distance for e-bikes, e-scooters, and bicycles are 1.46 m, 1.56 m, and 1.59 m, respectively. The average lateral distance for bicycles is found to be the largest, whereas e-bikes have the smallest lateral distance. This may be attributed to two reasons. The first is that bicycles are more willing to avoid lateral interference with automobiles by keeping a large lateral distance at the shared road where the bicycles, e-bikes, and e-scooters are mixed. Second, e-bikes and e-scooters are moving faster and bicycles tend to stay on the right hand side of the road, allowing space on the left hand side of the road for e-bikes and e-scooters.

The ANOVA test is conducted to test the significance in the difference of lateral distance for e-bikes, e-scooters, and bicycles, resulting in an F value of 3.162 with $p = .043$. The result indicates a significant difference of lateral distances for different two-wheelers classes. It is also found from the box plot Figure 4 that bicycles have the extreme lateral distance (the largest value and the smallest value), indicating that the bicycle riders show greater flexibility than e-bike and e-scooters riders when overtake automobiles. Furthermore, the average lateral distance for all types of two-wheelers is substantially greater than one meter.

5.2. Lateral distance distribution

To identify the theoretical distribution of the lateral distance, four commonly used distributions were examined in this study: Normal distribution, Lognormal distribution, Gamma distribution, and Weibull distribution. The

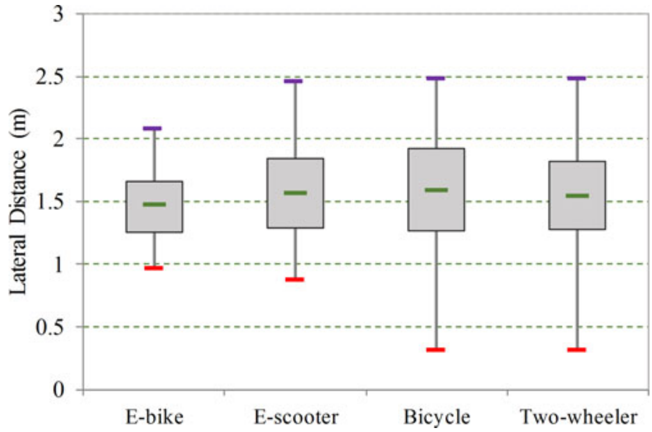

Figure 4. Analysis of the differences in lateral distance.

maximum likelihood estimation (MLE) technique was utilized to estimate the parameters of the distributions. The goodness-of-fit test was conducted to select the best-fitted distribution among the given candidate distributions. The Kolmogorov–Smirnov (K–S) test has been widely used to compare a sample with a particular probability distribution in transportation research (Ibeas, Cordera, dell'Olio, & Moura, 2011; Páez, Trépanier, & Morency, 2011). With the null hypothesis that the data follows a specified distribution, the K–S statistic estimates the largest vertical difference between the theoretical and the empirical cumulative distribution function (Washington et al., 2003). The null hypothesis is rejected at the chosen significance level if the test statistic is greater than the certain critical value. In this study, the K– test was conducted to test the hypothesis whether the lateral distance followed a particular probability distribution. The null hypothesis is accepted at a $(1 - \alpha)\%$ level of confidence if

$$D_n \leq K_\alpha/\text{square}(n) \tag{9}$$

Where, n is the samples size; D_n is the K–S statistic; K_α is the critical value (CV); $\alpha = .05$ in this study.

In the four candidate distributions, the distribution of the lowest K–S statistic is regarded as the best-fitted distribution. Table 1 shows the K–S test results, along with the parameters of the distribution. According to Table 1, several findings can be made:

1. For all the two-wheelers (the total data set), the Gamma distribution has the lowest K–S statistic (K–S = .032). Lognormal distribution also performs very well for the lateral distance (relatively small K–S values, K–S = .033). In reality, the two distributions have similar patterns. Taking the average lateral distance 1.542 m as an example: $P(G(16.719, 0.092) \leq 1.542) = .533$ and $P(\text{Lognormal}(0.403, 0.253) \leq 1.542) = .546$.

Table 1. Kolmogorov -Smirnov test of the lateral distance.

	Normal distribution			Lognormal distribution			Gamma distribution			Weibull distribution			
	Parameters	K–S[a]	p^b	Parameters	K–S	p	Parameters	K–S	P	Parameters	K–S	p	CV
E-bike	(1.47, 0.28)	0.060	0.84	(0.37, 0.19)	0.058	0.87	(28.48, 0.05)	0.056[a]	0.89	(1.58, 5.74)	0.082	0.48	0.13
E-scooter	(1.56, 0.35)	0.062	0.60	(0.42, 0.23)	0.070	0.43	(20.03, 0.08)	0.061[a]	0.60	(1.70, 4.97)	0.076	0.33	0.11
Bicycle	(1.59, 0.46)	0.087	0.42	(0.41, 0.33)	0.079	0.54	(10.37, 0.15)	0.057[a]	0.89	(1.76, 3.86)	0.091	0.37	0.14
Total	(1.54, 0.37)	0.064	0.11	(0.40, 0.25)	0.033	0.84	(16.72, 0.09)	0.032[a]	0.78	(1.69, 4.55)	0.073	0.05	0.07

[a]The K–S statistical of the best-fitted distribution.
[b]p is the probability of acceptance of the null hypothesis.

Figure 5a depicts the histogram and theoretical distributions for data samples with the Gamma and Lognormal distributions. As shown in the figure, the difference between the two distributions is very marginal.

2. For different two-wheelers classes, the lateral distance for e-bikes, e-scooters, and bicycles follows the Gamma distribution which is the best-fitted distribution. However, the shape parameters of the best-fitted Gamma distributions are within a relatively large range from 10.372 to 28.479. Figures 5b–d shows the histograms and theoretical distributions for data samples with the best-fitted distributions.
3. According to parameters of the best-fitted Gamma distribution, the cumulative distribution function (CDF) is calculated. For the lateral distance of all two-wheelers, 10% of values of the distribution is below 1.1 m (i.e., 10th percentile lateral distance), indicating that most of the lateral distance exceed 1.1 m. As such, this value (1.1 m) is defined as the critical lateral distance used in the following analysis.

5.3. Factors related to critical lateral distance

The FBBL model is estimated to evaluate the factors affecting the decision of a two-wheeler rider to accept small lateral distances. Critical lateral distance L_D is considered to be 1.1 m based on the distributional characteristics shown in the previous section. Namely, the $L_D = 1.1$ m is considered

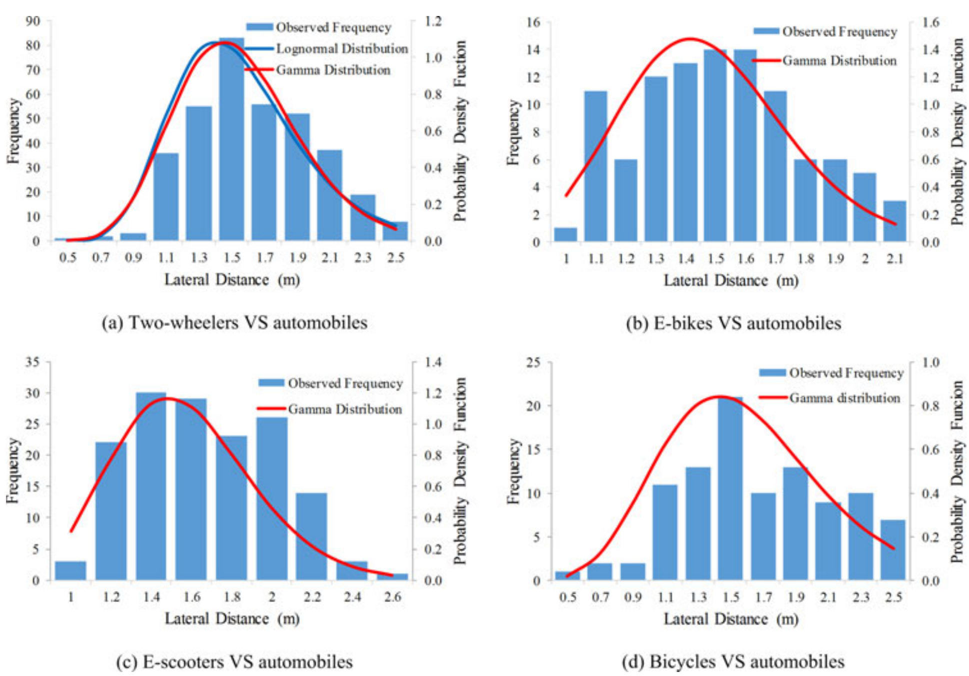

Figure 5. Distributions for lateral distance between two-wheelers and automobiles.

Table 2. Variables considered in the analysis of the two-wheelers traffic patterns.

Variables	Description
TW	Two-wheelers type: 1 = e-bike; 2 = e-scooter; 3 = bicycle
AU	Automobile type: 1 = Heavy vehicle; 0 = passenger car
LN_t	Lane number for two-wheeler: 1 = outside lane; 2 = middle lane; 3 = inside lane
LN_a	Lane number for automobile: 1 = middle lane; 0 = inside lane
EA	Evasive action: 1 = if two-wheeler takes evasive action; 0 = if automobile takes evasive action
PL_t	Existence of platoon of moving two-wheelers: 1 = yes; 0 = no
YAW_t	Yaw rate ratio of the two-wheeler (rad/s^2)
S_t	Speed of the two-wheeler (km/h)
$JERK_t$	Jerk of the two-wheeler (m/s^3)
YAW_a	Yaw rate ratio of the automobile (rad/s^2)
S_a	Speed of the automobile (km/h)
$JERK_a$	Jerk of the automobile (m/s^3)
SD	Difference of speed between the interacting two-wheeler and automobile (km/h)

as the cutoff point (marked as 1 if $L_D \leq 1.1\,\text{m}$; otherwise marked as 0) in the FBBL model. Candidate variables considered in the model are shown in Table 2. Prior to the FBBL model estimation, a correlation analysis is conducted to exclude correlated variables from further analysis.

The Bayesian analysis software WinBUGS (Lunn, Thomas, Best, & Spiegelhalter, 2000) is used as the modeling platform for model estimation. The Markov Chain Monte Carlo (MCMC) technique is used to approximate the probability distribution (i.e., the posterior Mean and Standard Deviation) of the parameters. Two independent Markov chains for each of the parameters run for 20,000 iterations. The first 10,000 interactions were used for monitoring convergence and then excluded as a burn-in sample. The remaining interactions were adopted for parameters estimation. To check convergence, two parallel chains with various starting values were tracked so that full coverage of the sample space was ensured. Brooks–Gelman–Rubin (BGR) statistic was also checked, where convergence occurs if the value of the BGR statistic is less than 1.2 (El-Basyouny & Sayed, 2009). Moreover, convergence was checked by visually inspecting the MCMC trace plots of the model parameters.

The results of the FBBL model is given in Table 3. Only the variables that are significant at a 5% level of significance are included (i.e., the ranges do not include a value with a sign different from the Mean). The model that has the best statistical fit is considered as the final form. Finally, six variables are found to be significantly associated with two-wheelers accepting the critical lateral distance. Figure 6 shows the posterior distributions for the significant variables. The OR is estimated to evaluate the quantitative impact of the interested explanatory variables on the probability of the outcome.

The type of two-wheelers is found to be a significant variable. The ORs for the e-bike and e-scooter binary variables are 3.487 and 4.337, respectively, indicating that e-bike and e-scooter riders are 3.487 and 4.337 times more likely, respectively, to accept the critical lateral distance than bicycle

Table 3. Estimation results of the full Bayesian binary logit (FBBL) model.

	Parameter estimates				ORs (Odds ratio)			
Variable	Mean	Standard deviation	2.5%	97.5%	Mean	Standard deviation	2.5%	97.5%
Intercept	−3.022	0.602	−4.284	−1.897	–	–	–	–
TW (Two-wheelers type): E-bike vs bicycle	1.130	0.481	0.222	2.119	3.487	1.858	1.249	8.323
TW (Two-wheelers type): E-scooter vs bicycle	1.363	0.448	0.535	2.297	4.337	2.158	1.707	9.949
AU(Automobile type): heavy vehicle vs passage car	−0.770	0.376	−1.532	−0.063	0.496	0.187	0.216	0.939
EA (Evasive action): two-wheelers vs automobiles	0.713	0.342	0.063	1.401	2.165	0.774	1.065	4.060
PL_t (Existence of platoon of moving two-wheelers): yes vs no	0.980	0.327	0.334	1.626	2.809	0.940	1.397	5.081
YAW_t (Yaw rate ratio of the two-wheeler)	5.953	2.991	0.112	11.840	–	–	–	–
SD (Difference of speed between the interacting two wheeler and automobile)	−0.042	0.025	−0.093	−0.006	–	–	–	–

— represents no value available.

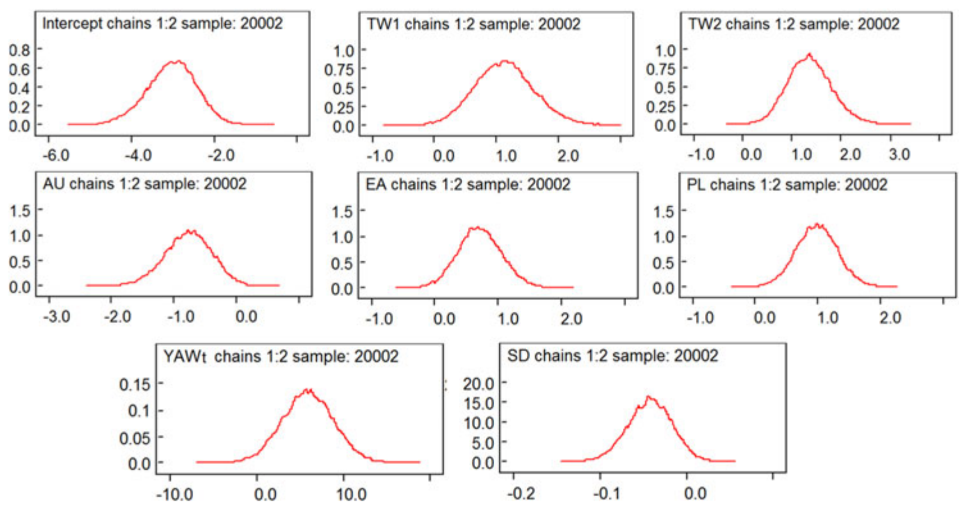

Figure 6. Posterior distributions for parameters of the significant variables.

riders are. The result is consistent with the comparative analysis of lateral distance in Figure 4 which shows that bicycles prefer larger lateral distance than e-bike and e-scooter riders. The result is logical as e-bikes and e-scooters are more likely to occupy the automobile lane to speed up or over-take. Furthermore, the OR for the e-scooter binary variable is larger than that for the e-bike binary variable, indicating that e-scooters are more relatively aggressive in accepting smaller lateral distance.

The presence of heavy vehicles is negatively related to the probability of two-wheeler riders accepting the critical lateral distance. The finding is consistent with the previous study by Sando et al. (2011) and Vlahogianni

(2014). A possible explanation for this can be related to the two-wheelers riders' perception of the higher risk for heavy vehicles. The OR for this binary variable is 0.496, indicating that when the interacting automobile is a heavy vehicle, the likelihood of two-wheeler rider accepting the critical lateral distance is 0.496 times of that when the interacting automobile is a passenger car.

The evasive action is significantly associated with the likelihood of two-wheelers accepting the critical lateral distance. In a lateral interference event, either the two-wheelers or the automobile takes evasive action to avoid the interference. The result of OR analysis indicates that when two-wheeler riders take evasive action, the likelihood of the two-wheeler rider accepting the critical lateral distance is 2.165 times more than that when automobiles take an evasive maneuver to avoid the lateral interference. The result suggests that the relationship of the lateral distance to the difference in sensitivity to the lateral interference between two-wheelers and automobiles. The finding indicates that the automobiles are more likely to avoid potential conflict at a relatively larger lateral distance compared to two-wheelers who accept smaller lateral distance. It also indicates that automobile drivers show higher sensitivity to the lateral interaction than the two-wheeler riders.

The existence of a platoon of moving two-wheelers is positively related to the likelihood of two-wheeler accepting the critical lateral distance. The OR for this binary variable suggests that when a platoon of moving two-wheelers exists, the probability of two-wheeler rider accepting the critical lateral distance is 2.809 times more than that when there is no platoon. The finding agrees with the fact that when there is a platoon of two-wheelers passing cars, some two-wheelers are moving faster and some are slower. The faster ones (e-bikes and e-scooters) are passing closest to cars. Field observation also reveals that two-wheeler riders occupy the adjacent motor lane and move close to automobiles to avoid further conflicts with other two-wheelers coming from the platoon.

The yaw rate refers to the angular velocity of the road-user rotation around the z-axis or the rate of change of the heading angle. The rotation describes the movement around the yaw axis of a rigid body that changes the direction to the left or right of its direction of motion. The model result shows that the probability of two-wheeler accepting the critical lateral distance increases with the yaw rate ratio of two-wheelers. As the above-mentioned that yaw rate ratio is an indicator for measuring the road-user swerving maneuvers, the finding can be explained that two-wheelers with larger yaw rate ratio take flexible swerving action when interacting with automobiles, which makes them accept shorter lateral distance.

The difference in the speed between the interacting two-wheeler and automobile is negatively related to the probability of two-wheeler rider accepting the critical lateral distance. The result is consistent with several previous studies (Stewart & McHale, 2014; Vlahogianni, 2014).

6. Conclusion and discussion

This study evaluated the lateral distance between overtaking two-wheelers and automobiles at a shared traffic road. Using a video-based computer vision technique, road users were tracked and grouped by a 3-D bounding boxes as an approximate model to collect the lateral distances. Lateral distance for different types of two-wheelers were examined and compared. Several lateral distance distributions were examined for different two-wheelers classes. Further, a full Bayesian binary logit model was developed to evaluate factors associated with the probability of two-wheelers accepting the critical lateral distance.

The following conclusions can be made from the analysis:

- The average lateral distance between overtaking two-wheelers and automobiles was 1.542 m.
- The average lateral distance for bicycles was the largest, followed by e-scooters and e-bikes. In addition to the significant difference in lateral distance for different two-wheeler classes, the lateral distance for all two-wheelers was generally greater than one meter.
- The K–S test showed that Gamma distribution was the best-fitted distribution of the lateral distance for all two-wheelers classes. The distributional analysis of the lateral distance resulted in a critical lateral distance of 1.1 m.
- The Two-wheelers type, presence of heavy vehicles, evasive action manner, occurrence of platoon of moving two-wheelers, two-wheelers' yaw rate ratio, and the speed difference between the interacting two-wheeler and automobile were significant in determining the likelihood for two-wheeler riders to accept the critical lateral distance.

Findings from this study can be beneficial for several transportation applications. The critical lateral distance derived from empirical data can be used to provide a theoretical basis for designing driving assistance systems. The statistical modeling results can be used to improve microsimulation models, in which two-wheelers' features, behavior, and interactions with automobiles can be enhanced with more realistic representation. Findings may also be used in two-wheelers safety studies for the development of countermeasures focusing on alleviating the factors related to the likelihood of accepting critical lateral distance. For example, e-bikes and e-scooters were found to be more likely to accept smaller lateral distance than a bicycle. This indicates that e-bikes and e-scooters may be more conflict prone than bicycles. As such, regulations or on-site enforcement should be developed to e-bikes and e-scooters. Moreover, the two-wheelers' swerving maneuver that is reflected by the yaw rate is significantly related to the likelihood of accepting critical lateral distance. Therefore, local laws could be proposed to regulate their riding behaviors. Furthermore, the 3-D model used for detecting the lateral distance shows a scalable application of computer vision techniques. This model improves the quality of road users' tracking. It also enables a wider range of road user classes identification.

There are some limitations of this study. The data used in this study was limited to only one shared traffic road. Two-wheelers' driving patterns vary among different countries, cultural contexts, and traffic environments. Therefore, future work should expand this study to include more shared traffic facilities using a extensive video data set. With the larger data set, additional valuable variables such as road widths, the number of lanes, parking condition, and different speed limits can be incorporated into the statistical model. Finally, more work is required on investigating the relationship between conflicts, collisions and the lateral interferences.

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